Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/03/2022
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (80NSSC19K0231 ; NNX16AK21G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	07/01/2016	**End Date:**	06/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2022 per PI (Ed., 8/5/21) NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/12/20) NOTE: Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and MSFC personnel (Ed., 2/21/19)		
Key Personnel Changes/Previous PI:	Dr. Elisa Torresani took over student supervision starting in the summer of 2020 and is guiding both characterization and fabrication efforts as Co-Investigator. She is supervising one PhD student at this time.		
COI Name (Institution):	Olevsky, Eugene Ph.D. (San Diego State University) Torresani, Elisa Ph.D. (San Diego State University)		
Grant/Contract No.:	80NSSC19K0231 ; NNX16AK21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 2/21/19): Extended to 6/30/2020 with new grant number 80NSSC19K0231 per NSSC and Marshall Space Flight Center (MSFC) personnel. NOTE (Ed., June 2016): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments," grant #NNX14AB31G with the same Principal Investigator. A flight research project is in progress investigating the gravity-dependent densification, distortion, microstructure development, and the related macroscopic shape evolution of multi-phase powder alloys during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the liquid phase sintering induced pore-grain structure evolution by the de-convolution of the gravity impact and (ii) exploring sintering under microgravity conditions as the basis for techniques applied to in-space fabrication and repair via additive manufacturing. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit particle-based fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing based on sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer finite element code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity are being performed using a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon. As of April 2022 all microgravity experiments are completed, consisting of 49 samples. As of March 2022, the last 14 microgravity samples were returned to the research team. These last 14 samples are under current analysis for dimensional change, distortion, densification, and microstructure evolution. At the same time, based on the microgravity findings, new samples are under fabrication for ground proof experiments. A total of 14 samples, representing 2 cartridges, are sought following the C2 and C5 heating protocols. As of March 2022, all samples from the International Space Station are in hand. Ground proof experiments are not scheduled as yet, but sample fabrication is underway (the original ground proof samples were lost during the pandemic due to a death from a support engineer in Huntsville). So in short three prongs of activity are underway - 1) ground proof sample fabrication, 2) microgravity sample characterization, 3) behavior modeling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, magnets, and composites. In the USA this is responsible for approximately $25 billion in annual component production. The products are applied to such diverse fields as metal cutting, oil well drilling, automotive engines, electric motors, computer heat sinks, jet engines, medical tools, skeletal implants, exercise monitors, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge in sintering since often dimensions change 15 to 20% due to shrinkage in going from the porous powder structure to the dense component. Gravity provides for a nonuniform shrinkage in height versus perpendicular directions. During high temperature sintering the component strength drops considerably and is subject to gravity induced settling effects while pores undergo buoyancy. Removing gravity allows surface energy to spheroidize both the component and pore space (agglomerated pores form a massive void). Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a verified computational model to enable prediction of final size and shape for different gravitational fields, using a multiple scale model with hopes to extend from geometry to property predictions. The robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions, as required for eventual additive manufacturing as potentially envisioned for fabrication on the moon, other planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles. Of course, this has benefit on Earth with the evolution of a computational means to predict high tolerance shapes by guiding sintering practice. We are adding the missing gravitational term to computer simulations, typically finite element analysis tools, for predicting final shape and size. New insight has been gained on the role of pore saturation in component shape loss, indicating that under-sintering in reduced gravity is the appropriate protocol for net-shape fabrication. A mathematical model embedded in finite element analysis is offered to predict system softening versus densification.
Task Progress:	This project focuses on sintering; it conceptually examines liquid phase sintering. The solid phase is tungsten and the liquid is a molten alloy of copper-nickel-manganese with some dissolved tungsten. The pores initially constitute 37 % of the sample volume and the liquid to solid ratio is varied by changing the tungsten content. The lowest tungsten is 70 wt %, corresponding to approximately 50 to 55 vol. % solid (deepening on the liquid composition and tungsten solubility in the liquid). The highest tungsten content is 90 wt % or about 80 vol. % solid. Three matrix or liquid alloys are formed from different ratios of copper, nickel, and manganese. The matrix melts between 1145 and 1185 deg C. Microgravity sintering experiments are completed, involving seven cartridges with seven samples each. Two temperatures are used, 1180 and 1210 deg C. Hold time at the lower temperature is short, capturing the liquid formation transient. Times at the higher temperature range up to 90 min in steps of 10, 13, 20, 35, and 69 min of isothermal hold. The flight hardware then captures seven samples for each run, reflecting different liquid to solid ratios and different matrix alloys with different wetting conditions. Upon return, the samples are subjected to a battery of first nondestructive tests (photographs, mass, diameter, density by immersion, distortion by coordinate measuring machine, and surface examination of large pores, defects, cracks, or other abnormal features). The samples are then cross-sectioned for microstructure. Gradients in grain size and solid-liquid fraction are measured. Grain size is based on linear intercept measurements. The grain size distribution is fit to a Weibull distribution that is reflective of the microstructure. Pores are evident after sintering and both the size and location of the large pores are measured. The time-temperature cycle is integrated to provide a parameter known as the work of sintering and that is used to evaluate for systematic densification. In early ground experiments, samples were in contact with the containment crucibles, leading to adhesion. In the flight experiments, where there was no gravity induced contact, no sticking was observed. Relative density after sintering was

	high for the low solid content alloys even after short hold times. The 70 wt % tungsten sample reached 93.6 % density for the 1180 deg C run, while the 90 wt % tungsten only reached 70.6 % density in the same run. Shrinkage in sintering ranged from 0 % to over 12 % depending on the sample. Diameter shrinkage was similar. Mass loss generally was small, and most typically associated with manganese evaporation. Most curious was the fact that the samples showed evidence of no densification, for the most part, once a density near 80 to 90 % was achieved. The lack of buoyancy force resulted in pore agglomeration, coalescence, and expansion. Such factors are not typically encountered in sintering since gravity causes the compression and densification of the pore space. In microgravity, the pores are only impacted by surface energy, and tend to minimize surface area by forming spheres, but do not annihilate as typically seen on Earth.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
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